

Analytical chemistry

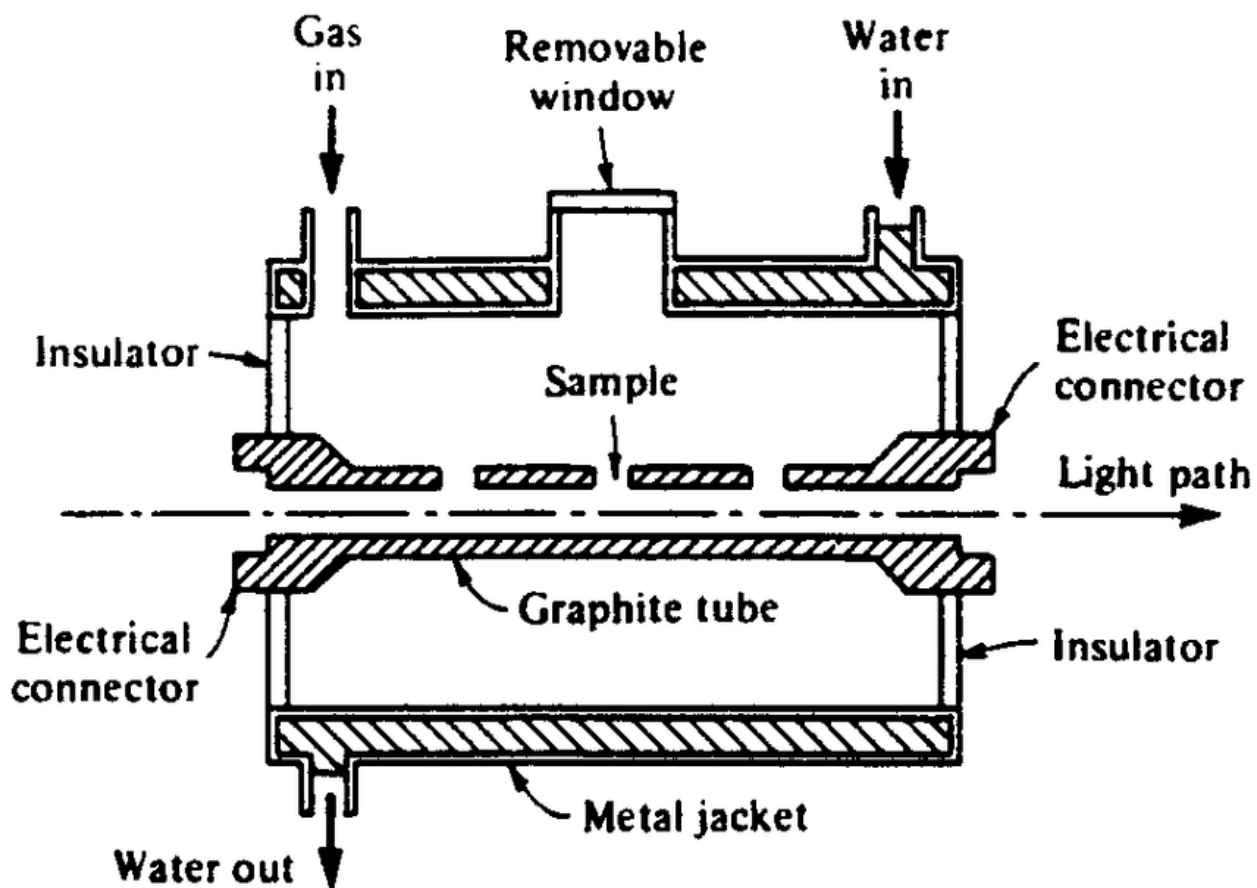
(CHEM-319)

Long Questions:-

(1)

Explain the construction and working of graphite furnace?

A graphite furnace, also known as a graphite tube furnace or electro-thermal atomizer, is a device used in analytical chemistry and atomic absorption spectroscopy to determine the concentration of various elements in a sample. It works by heating a sample to a high temperature within a graphite tube and then measuring the amount of light absorption at a specific wavelength, which is proportional to the concentration of the element of interest.



Construction:

Graphite Tube: The central component of the graphite furnace is a small, cylindrical tube made of high-purity graphite. This tube serves as the sample container and also plays a critical role in heating the sample to a high temperature.

Furnace Cavity: The graphite tube is inserted into a furnace cavity, which provides a controlled and stable heating environment. The cavity is typically made of a refractory material that can withstand high temperatures.

Heating Elements: Surrounding the cavity are heating elements, such as resistive coils or electrical heating elements. These heating elements provide the necessary energy to heat the graphite tube and the sample.

Gas Inlet and Outlet: There are gas inlets and outlets connected to the graphite furnace. The gas inlet allows for the introduction of various gases, such as an inert gas (usually argon) or a modifier gas. The gas outlet helps remove unwanted gases and combustion products.

Working:

Sample Introduction: A small volume of the liquid sample containing the element to be analyzed is introduced into the graphite tube. The sample is typically injected using a syringe or an auto-sampler.

Drying and Ashing: Initially, the temperature is raised to evaporate any solvent or moisture from the sample. This step is known as drying. After drying, the temperature is increased further to ash any organic material, leaving behind the inorganic components.

Atomization: The temperature is then rapidly increased to a higher level, causing the sample to vaporize and atomize. This process converts the analyte into free atoms in a gaseous state. The atoms are usually in the ground state.

Light Absorption Measurement: A hollow cathode lamp or other light source emits a specific wavelength of light that corresponds to the element of interest. The atoms in the vaporized sample absorb this light at the resonance wavelength. The amount of absorption is directly proportional to the concentration of the element in the sample.

Data Analysis: The amount of absorbed light is measured and recorded by a detector, such as a photomultiplier tube or a photodiode array. The resulting data is then analyzed to determine the concentration of the element in the sample.

Conclusion:

The graphite furnace's ability to atomize and analyze trace elements with high sensitivity makes it a valuable tool in analytical chemistry and elemental analysis. The controlled heating process within the graphite tube and the precise measurement of light absorption allow for the quantification of even very low concentrations of elements in a wide range of sample types.

(2)

Discuss different components or steps of flow injection analysis?

Flow Injection Analysis (FIA) is an automated analytical technique used for the rapid and precise analysis of a wide range of chemical and biochemical samples. FIA systems consist of various components that work together to perform sample handling, reagent addition, mixing, and detection.

Components/Steps:-

Here are the key components of a typical FIA system:.

1. Sample Injection System:

Sample Injection Valve: *This valve is responsible for introducing a precise and reproducible volume of the sample into the FIA system. It can be a rotary injection valve or a multi-position valve.*

Sample Loop: *The sample loop is a fixed-volume chamber within the injection valve that holds a known volume of the sample. The sample is loaded into the loop before injection.*

2. Carrier Fluid (Reagent) Delivery System:

Carrier Fluid Reservoir: *This container holds the carrier fluid (often a buffer or solvent) that is used to transport the sample and reagents through the system.*

Carrier Fluid Pump: *A peristaltic or syringe pump delivers the carrier fluid at a controlled flow rate. The pump ensures a consistent and reproducible flow of the sample and reagents through the system..*

3. Sample/Reagent Mixing and Reaction Coil:

Coiled Tubing or Mixing Chamber: *This component allows for thorough mixing of the sample and reagents. In some cases, it may also serve as a reaction coil, where chemical reactions can take place before detection.*

4. Reagent Delivery System:

Reagent Reservoirs: *Containers that hold reagents and standards required for the analysis.*

Reagent Pump: A pump delivers reagents from the reservoirs at a controlled flow rate. Reagents are merged with the sample stream in a precise manner.

5. Detector:

Detector Cell: This is the site where the analytical signal is generated. It may contain a specific sensor, such as a photometric cell, fluorescence cell, or electrochemical cell, depending on the type of analysis being performed.

Signal Transducer: The signal generated by the detector cell is typically converted into an electrical signal using a transducer. The choice of transducer depends on the detection principle used (e.g., photodiode for photometric detection, amperometric cell for electrochemical detection).

Signal Processor: The electrical signal is processed and recorded by a computer or data acquisition system. This may involve data smoothing, integration, and calibration.

6. Waste Management System:

Waste Container: FIA systems often generate waste in the form of used carrier fluid, sample, and reagents. A waste container collects and disposes of this waste.

7. Control and Data Acquisition System:

Computer or Controller: This component controls the operation of the FIA system, including sample injection, carrier fluid flow, reagent addition, and data acquisition.

Data Acquisition Software: Specialized software is used to collect, process, and analyze data generated by the detector.

8. Safety and Monitoring Systems: Depending on the specific application, safety features such as pressure sensors, leak detectors, and emergency shut-off mechanisms may be integrated into the FIA system.

(3)

Write a detail note on types of resins in Ion Exchange chromatography?

Ion exchange chromatography is a widely used separation technique in analytical and preparative chemistry. It separates ions based on their charge by utilizing ion exchange resins. These resins contain fixed ionic groups that attract and retain ions of the opposite charge. There are different types of ion exchange resins, each with unique properties, making them suitable for various applications. Here are the primary types of resins used in ion exchange chromatography:-

Cation Exchange Resins:

Cation exchange resins are a type of ion exchange resin that contains fixed negatively charged functional groups, typically in the form of sulfonic acid (-SO₃H) or carboxylic acid (-COOH) groups. These resins are designed to selectively exchange cations (positively charged ions) in a sample solution for hydrogen ions (H⁺) or other cations present in the resin.

Strong Acid Cation (SAC) Resins: *These resins contain sulfonic acid groups (-SO₃H) and are used for the separation of cations. They offer high acidity and are effective for strongly acidic conditions.*

Weak Acid Cation (WAC) Resins: *WAC resins have carboxylic acid groups (-COOH) and are used for the separation of cations. They have lower acidity than SAC resins and work better under mildly acidic conditions.*

Anion Exchange Resins:

Anion exchange resins are a type of ion exchange resin that contains fixed positively charged functional groups, typically in the form of quaternary ammonium (-NR₄⁺) or primary, secondary, or tertiary amine groups. These resins are designed to selectively exchange anions (negatively charged ions) in a sample solution for other anions, often hydroxide ions (OH⁻) or other negatively charged ions present in the resin.

Strong Base Anion (SBA) Resins: *SBA resins feature quaternary ammonium groups (-NR₄⁺) and are used for the separation of anions. They provide strong basicity and are suitable for strongly basic conditions.*

Weak Base Anion (WBA) Resins: *WBA resins have primary, secondary, or tertiary amine groups and are used for the separation of anions. They have weaker basicity than SBA resins and are suitable for mildly basic conditions.*

(4)

Describe hydride generation in flameless methods in AAS?

Hydride generation is a technique used in flameless atomic absorption spectrometry (AAS) to improve the sensitivity and selectivity of the analysis, especially for the determination of certain elements like arsenic (As), antimony (Sb), and selenium (Se) that do not readily atomize and absorb in a flame. The process of hydride generation involves the generation of volatile hydrides of these elements, which are then introduced into the AAS instrument for measurement. Here's how hydride generation works in flameless AAS:

Sample Preparation:

The sample to be analyzed is typically in the form of a solution. It may be a water sample, biological fluid, or any other liquid matrix. The sample may contain the element of interest, along with other potentially interfering substances.

Hydride Generation Reaction:

To convert the non-volatile element into a volatile hydride, a reducing agent, such as sodium borohydride (NaBH_4) or stannous chloride (SnCl_2), is added to the sample. This reducing agent reacts with the element to form the corresponding hydride gas. For example, arsenic forms arsine (AsH_3), antimony forms stibine (SbH_3), and selenium forms hydrogen selenide (H_2Se).

Hydride Trap:

The generated hydride gas is directed into a hydride trap, which is typically a cold trap. This trap is usually cooled with liquid nitrogen or another cryogenic coolant to condense and collect the volatile hydrides.

Transport to the Atomization System:

After trapping the hydrides, the trap is heated to release the condensed hydrides as a gaseous mixture.

Introduction into the Flameless Atomizer:

The gaseous mixture of hydrides is then introduced into the atomization system of the flameless AAS instrument. This system typically consists of a graphite furnace or an electro-thermal atomizer. The hydrides are subjected to high temperatures in the atomizer, causing them to thermally decompose, releasing the target element in atomic form.

Atomization and Absorption:

The atoms of the target element are then introduced into the light path of the AAS instrument. A monochromatic light source, such as a hollow cathode lamp, emits light at a specific wavelength that corresponds to the absorption characteristics of the element. The atoms absorb the light at the resonance wavelength, and the amount of light absorbed is proportional to the concentration of the element in the sample.

Signal Detection and Analysis:

A detector in the AAS instrument measures the intensity of the absorbed light, and the resulting signal is recorded. The instrument's software processes the signal and calculates the concentration of the element in the sample.

(5)

Give an account of capillary zone electrophoresis?

Capillary zone electrophoresis (CZE) is a high-resolution separation technique used for the analysis of charged analytes, such as ions, small molecules, and biomolecules. It is a form of capillary electrophoresis (CE) that takes place in a narrow fused-silica capillary tube. CZE is based on the differential migration of charged species in an electric field through a buffer-filled capillary. Here is an account of the key aspects of capillary zone electrophoresis:

Basic Principles:

Capillary Tube: CZE is conducted in a fused-silica capillary tube with an inner diameter typically in the range of 25-100 micrometers. The capillary is filled with an electrolyte buffer solution.

Electric Field: An electric field is applied across the capillary by placing one end in a cathode buffer and the other end in an anode buffer. The application of the electric field causes charged species to migrate within the capillary based on their charge-to-size ratio (electrophoretic mobility).

Separation Mechanism: Separation in CZE is based on differences in electrophoretic mobility of the analytes. Positively charged analytes migrate toward the cathode (negative electrode), while negatively charged analytes migrate toward the anode (positive electrode).

Key Components:

High-Voltage Power Supply: A power supply is used to generate the electric field applied to the capillary. The voltage and current are carefully controlled to optimize the separation.

Capillary: The fused-silica capillary is a critical component, as it provides a narrow separation channel, which enhances the separation efficiency. The inner surface of the capillary is often coated to reduce analyte-wall interactions.

Injection System: Samples are introduced into the capillary through an injection system. Common injection methods include electrokinetic injection, pressure injection, and hydrodynamic injection.

Detection System: Detection is typically performed using UV-Vis absorbance, fluorescence, or conductivity detectors. The capillary can be transparent for UV and fluorescence detection.

Advantages of CZE:

High Resolution: CZE offers excellent resolution, making it suitable for the separation of closely related compounds, including ions, small molecules, amino acids, and peptides.

Minimal Sample Requirements: CZE requires only small sample volumes, often in the microliter range, which is advantageous for applications with limited sample availability.

Capillary zone electrophoresis is a versatile and powerful separation technique that offers high resolution and is widely used in various fields of analytical chemistry for the separation and analysis of charged compounds.

(6)

Describe factors affecting ion exchange chromatography?

Factors affecting ion exchange separations

- **1. Nature & properties of ion exchange resins:-** Cross linking & swelling
- **2. Nature of exchanging ions:**
- **A) Valency of ions:** at low conc & ordinary temp , extent of exchange increases with increase in valency
$$\text{Na}^+ < \text{Ca}^{2+} < \text{Al}^{3+} < \text{Th}^{4+}$$

Factors affecting ion exchange separations

- **B) Size of ions:** for similar charged ions, exchange increases with decrease in the size of hydrated ion.
- $\text{Li} < \text{H}^+ < \text{Na}^+ < \text{NH}_4^+ < \text{K}^+ < \text{Rb}^+ < \text{Cs}^+$
- **C) Polarizability:** exchange is preferred for greater polarizable ion
Eg: $\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^-$
- **D) conc of solution:** in dilute solutions, polyvalent anions are generally adsorbed preferentially
- **E) conc and charge of ions:** if resin has higher +ve charge and solution has lower +ve charge , exchange is favoured at higher conc.
- If the resin has lower +ve charge and solution has high +ve charge , then exchange is favoured at low conc.

(7)

Explain Zeta Potential in detail on the basis of electro-osmotic flow?

Zeta Potential:

Zeta potential (ζ) is a measure of the magnitude of the electric potential at the shear plane, which is typically located very close to the particle's surface. It arises from the difference in charge between the particle's surface and the surrounding liquid. Zeta potential can be either positive or negative, indicating the presence of positive or negative surface charges on the particle. This charge distribution creates an electrical double layer around the particle, consisting of the inner Helmholtz layer (near the particle) and the outer diffuse layer.

The zeta potential (ζ) can be calculated from electrophoretic mobility (μ) using the Henry equation, which relates electrophoretic mobility to zeta potential:

$$\mu = \epsilon \zeta / \eta$$

Electro-Osmotic Flow (EOF):

Electro-osmotic flow is the motion of a liquid phase in a porous medium or a capillary under the influence of an applied electric field. In the context of electrophoresis or liquid chromatography, it refers to the movement of the mobile phase in a column, capillary, or microfluidic channel when an electric field is applied. EOF is typically induced by the surface charge on the walls of the medium and the interaction of this charge with the electric field.

The electro-osmotic flow rate (v) can be described by the Smoluchowski equation:

$$v = \epsilon \zeta E / \eta$$

Relationship Between Zeta Potential and Electro-Osmotic Flow:

The zeta potential of the channel or medium plays a crucial role in determining the magnitude and direction of electro-osmotic flow. The sign and magnitude of the zeta potential will affect the direction of fluid movement under the influence of an applied electric field. If the zeta potential is negative, as is often the case for silica or glass surfaces, the electro-osmotic flow will be in the direction of the cathode (negative electrode). Conversely, if the zeta potential is positive, the flow will be towards the anode (positive electrode).

(8)

Explain Counter-current process in solvent extraction?

Counter-current extraction is a commonly used technique in solvent extraction processes for separating or purifying components of a solution or mixture based on their differing solubilities in two immiscible phases. In counter-current extraction, two phases, typically a liquid phase (organic solvent) and an aqueous phase, are continuously circulated in opposite directions to maximize the transfer of the solute between the phases. This process is widely employed in chemical engineering and the chemical industry for various applications, including metal extraction, purification of organic compounds, and separation processes. Here's an overview of the counter-current extraction process:

Key Components and Steps in Counter-Current Extraction:

Extraction Equipment: The counter-current extraction process is typically conducted using a series of extraction stages or units, such as mixer-settlers or extraction columns. These units are designed to facilitate contact between the two immiscible phases.

Two Immiscible Phases:

- **Aqueous Phase:** This phase contains the solute or target component that needs to be separated or extracted. It is often referred to as the feed phase.
- **Organic Phase:** The organic phase contains the extracting solvent, which has an affinity for the solute of interest. The solute transfers from the aqueous phase to the organic phase during the extraction.

A simple type of apparatus scale is a **Craig apparatus**. Craig apparatus consists of a series of glass tubes (r: 0, 1, 2..) that are designed and arranged such that the lighter liquid phase is transferred from one tube to the next.



Counter-Current Flow: The key feature of counter-current extraction is that the two phases flow in opposite directions within the extraction units. The aqueous phase flows in one direction (e.g., from the top to the bottom), while the organic phase flows in the opposite direction (e.g., from the bottom to the top).

Multiple Stages: Counter-current extraction is typically conducted in multiple stages or units connected in series. This arrangement allows for the gradual separation of the solute as it progresses through the stages.

Phase Separation and Collection: After passing through the extraction units, the two phases are separated, often in a settling tank, and the organic phase, now enriched with the extracted solute, is collected for further processing or analysis.

(9)

Write down the working and characteristics of hollow cathode lamp?

A hollow cathode lamp (HCL) is a specialized type of discharge lamp commonly used in analytical chemistry instruments, such as atomic absorption spectrometers and inductively coupled plasma (ICP) optical emission spectrometers. Hollow cathode lamps are designed to emit narrow, intense lines of light at specific wavelengths, making them ideal for the quantitative analysis of elements in a sample. Here's an overview of the working and characteristics of a hollow cathode lamp:

Working of a Hollow Cathode Lamp:

Cathode and Anode: The hollow cathode lamp consists of a cylindrical, thin-walled tube with one end closed, serving as the cathode, and the other end open, serving as the anode. The cathode is typically made of the metal corresponding to the element of interest.

Rare Gas Filling: The lamp is filled with a rare gas, such as argon (Ar) or neon (Ne), at low pressure. The choice of gas depends on the application and the desired spectral lines.

Electrical Discharge: When an electrical potential (voltage) is applied across the cathode and anode, a discharge occurs in the gas within the lamp. This discharge ionizes the gas and creates a plasma within the lamp.

Radiative Transition: When the metal atoms or ions are excited by the discharge, they return to their ground state by emitting photons in the form of characteristic spectral lines. These lines correspond to specific wavelengths unique to the element being analyzed.

Light Emission: The emitted light exits the lamp through the open end. The lamp's design ensures that the emitted light is directed toward the entrance slit of the spectrometer, facilitating spectral analysis.

Characteristics of Hollow Cathode Lamps:

Monochromatic Light: Hollow cathode lamps emit intense, monochromatic (single-wavelength) light. Each lamp is designed to emit specific spectral lines corresponding to the

element of the cathode material. This mono-chromaticity is essential for accurate elemental analysis.

High Intensity: *HCLs produce highly intense lines of light, which are beneficial for achieving low detection limits in analytical instruments.*

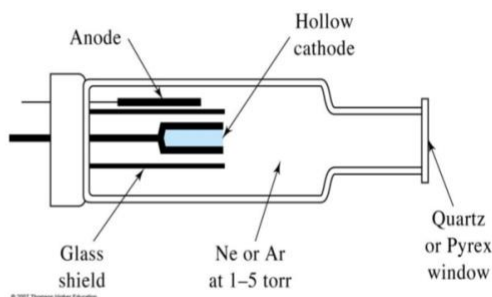
Longevity: *Hollow cathode lamps have a relatively long operational life, making them cost-effective and practical for routine use in analytical laboratories.*

Narrow Line Width: *The emitted spectral lines have a narrow line width, enhancing the instrument's selectivity.*

Hollow cathode lamps are essential components of atomic spectroscopy instruments and are widely used for the determination of trace elements in a variety of samples, including environmental, biological, and industrial samples.

Hollow Cathode Lamps:

- The most common source for AAS.
- Consists of a tungsten anode and a cylindrical cathode sealed in a glass tube that is filled with **neon or argon at a pressure of 1 to 5 torr.**
- The cathode is constructed of the metal whose spectrum is desired.



- Ionization of the inert gas occurs when a potential on the order of 300 V is applied across the electrodes, which generates a current of about 5 to 15 mA.
- If the potential is sufficiently large, the gaseous cation acquire enough kinetic energy to dislodge some of the metal atoms from the cathode surface and produce an atomic cloud in a process called **sputtering**.
- A portion of the sputtered metal atom are in excited states and thus emit their characteristic radiation as they return to the ground state.
- Eventually, the metal atoms diffuse back to the cathode surface or to the glass walls of the tube and are redeposited.



(10)

Briefly explain the principle, process and applications of gel chromatography?

Principle of Gel Chromatography:

Gel chromatography, also known as size exclusion chromatography (SEC), is a chromatographic separation technique based on the principle of separating molecules by their size and shape. It is a non-adsorptive chromatography method that relies on the differential exclusion of molecules from the pores of a porous gel matrix. Smaller molecules can enter the pores of the gel, resulting in longer and more convoluted paths through the column, while larger molecules cannot enter the pores and take shorter, direct paths.

Process of Gel Chromatography:

The process of gel chromatography involves the following steps:

Column Packing: *A column is packed with a gel matrix consisting of porous beads. The gel is chosen based on the desired separation range, and it typically has a specific pore size distribution.*

Sample Loading: *A mixture of molecules of interest is applied to the top of the column.*

Elution: *A mobile phase (often a buffer) is continuously introduced at the top of the column. As the mobile phase flows through the column, it carries the sample components down the column.*

Separation: *Larger molecules are excluded from the pores of the gel matrix and travel through the column more quickly because they take shorter paths. Smaller molecules enter the pores, resulting in longer and more tortuous paths, leading to delayed elution. The net effect is that molecules are separated based on their size, with larger molecules eluting earlier and smaller molecules eluting later.*

Collection: *Fractions of eluted material are collected and analyzed for the presence of specific molecules.*

Applications of Gel Chromatography:

Gel chromatography is widely used in various applications for the separation, purification, and analysis of biomolecules and other compounds. Some common applications include:

Protein Purification: *Gel chromatography is used to separate and purify proteins based on their molecular size, allowing for the isolation of specific protein fractions from complex mixtures.*

Polysaccharide Separation: *It is employed in the separation of polysaccharides, including carbohydrates and glycoproteins.*

Nucleic Acid Analysis: *Gel chromatography is used in the analysis and purification of nucleic acids, such as DNA and RNA.*

Quality Control: *It is commonly used in quality control and research applications to analyze and characterize complex mixtures of biomolecules or other compounds.*

Gel chromatography is particularly useful for separating and characterizing macromolecules, such as proteins, nucleic acids, and polysaccharides, making it an essential technique in biochemistry, biotechnology, and various other scientific fields. It is valued for its ability to provide size-based separation without the need for complex chemical interactions.